CLUSTER FOR CLEAN ENERGY AND CLIMATE CHANGE (CCECC), BOWEN UNIVERSITY, IWO.

Cluster for Clean Energy and Climate Change

1ST NATIONAL CONFERENCE ON CLEAN ENERGY AND CLIMATE CHANGE
15-16TH APRIL, 2026

CCECC PROCEEDINGS

Advancing Clean Energy Solutions for Climate-Resilient and Sustainable Development in Nigeria

NCCECC Proceedings **Vol 1, Pg 18-32**

Advances in Sustainable Energy Technologies Towards Net-Zero Emissions in Maritime Operations

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Abstract

The maritime shipping industry accounts for approximately 2-3% of global greenhouse gas emissions. The International Maritime Organization (IMO) has set a target to reduce Greenhouse Gas (GHG) emissions by at least 50% by 2050 compared to 2008 levels. The maritime sector faces unprecedented pressure to implement sustainable technologies while maintaining operational efficiency and economic viability. This current study presents a systematic review of advances in sustainable energy technologies for maritime decarbonization. The study scope covers alternative fuels, propulsion systems, energy-efficiency technologies, digital innovations, and policy mechanisms essential to achieving net-zero maritime operations. The review examines peer-reviewed literature and industry reports that synthesized evidence on technological maturity, environmental performance, economic feasibility, and implementation barriers across four technology categories. The four technology categories include: (1) alternative fuels, including LNG, methanol, ammonia, and hydrogen; (2) energy efficiency technologies encompassing hull optimization, waste heat recovery, and wind assistance; (3) electric and fuel cell propulsion systems; and (4) digital technologies. The study reveals that sustainable systems that incorporate digital technologies and AI-driven optimization enable an additional 3-5% in operational fuel savings. However, capital costs represent the dominant implementation barrier, with an investment of about $277-835 billion through 2050 across fuel infrastructure, vessel retrofits, and port electrification. The convergence of advancing technologies, declining renewable energy costs, and strengthening regulatory frameworks creates unprecedented opportunity for maritime decarbonization while maintaining the sector's essential role in global commerce.

Keywords: Greenhouse gas emissions, Energy-efficiency, Net-zero emissions, AI-driven optimization

1. Introduction

Maritime transport plays a fundamental role in global trade, accounting for over 90% of international trade by sea, making shipping the backbone of the global economy (Uchechukwu & Mba, 2024). However, this sector accounts for approximately 2-3% of global greenhouse gas (GHG) emissions, a figure comparable to aviation and significantly higher than that of many other industries (Bayer et al., 2025). Mallouppas & Yfantis (2021) noted that the International Maritime Organization (IMO) has identified shipping emissions as a critical contributor to climate change, with over 870 million tonnes of CO_2 emitted annually by international shipping. Beyond carbon dioxide, maritime operations emit substantial quantities of sulphur oxides (SOx), nitrogen oxides (NOx), and particulate matter (PM), which collectively pose severe health and environmental risks (Filippopoulos et al., 2024).

The rapid expansion of maritime trade, coupled with larger vessels and globalization, has intensified pressure on the shipping industry to adopt sustainable practices (Liu et al., 2024). Bogalecka et al. (2026) confirmed that climate change itself presents additional operational challenges for maritime transport, as rising sea levels, increased weather variability, and changing ocean currents create new uncertainties for shipping routes and operational costs. Consequently, the maritime industry faces a dual imperative: reducing its environmental footprint while adapting to a changing climate. Recognizing the urgency of decarbonization, the IMO adopted the GHG Strategy in 2018, setting ambitious targets to reduce greenhouse gas emissions from international shipping (Uchechukwu & Mba, 2024). Bayraktar et al. (2025) identified the industry's initial strategy to reduce total GHG emissions from shipping by at least 50% by 2050. Hansson et al. (2020) discussed that the framework represents the first legally binding international agreement specifically addressing shipping emissions and aligns with the Paris Agreement's goal to limit global temperature rise to well below 2°C. There is a need to further explore the trend and the impact of sustainable technology on emission reduction in maritime operations.

The current study systematically examines the multifaceted advances in sustainable energy systems for marine operations. The study explores four technologies towards net-zero emissions in the maritime industry. The study explores: (1) alternative fuels including LNG, methanol, ammonia, and hydrogen; (2) energy efficiency technologies encompassing hull optimization, waste heat recovery, and wind assistance; (3) electric and fuel cell propulsion systems, and (4) digital technologies including Internet of Things (IoT), artificial intelligence, and autonomous systems. The study reveals that sustainable systems that incorporate digital technologies and AI-driven optimization enable an additional 3-5% in operational fuel savings. The convergence of advancing technologies, declining renewable energy costs, and strengthening regulatory frameworks creates unprecedented opportunity for maritime decarbonization while maintaining the sector's essential role in global commerce.

2. Literature Review

This section provides a brief overview of the literature on advances in alternative fuels and energy carriers, as well as on transitioning away from conventional fuels.

2.1. Liquefied Natural Gas (LNG) as a Transition Fuel

Uchechukwu & Mba (2024) explore LNG as an emerging, immediately viable alternative fuel for maritime applications, with substantial infrastructure development already underway globally. Maha-Prados et al. (2026) confirmed that LNG offers significant advantages over conventional heavy fuel oil. This includes reduced CO_2 emissions of approximately 15-20%, near-complete elimination of Sulphur emissions, and reductions in particulate matter of up to 90% (Maha-Prados et al., 2026). The relatively mature technology for LNG production and distribution, combined with existing experience in the liquefied gas sector, has facilitated rapid adoption among shipping companies seeking to comply with regulatory requirements (deManuel-Lpez et al., 2024).

However, the environmental benefits of LNG are constrained by methane slip, unburned methane released during engine operation, which significantly undermines the fuel's climate impact when considering full lifecycle emissions (Bayraktar et al., 2025). Melnyk et al. (2025)

show that LNG can provide short-term emission reductions but faces limitations as a long-term decarbonization solution due to its fossil-fuel origin and concerns about methane leakage. Alexander & Friday (2024) explore the supply chain for LNG and its associated infrastructure, and identify that associated challenges in many ports, particularly in developing regions, limiting access for certain vessel types and trade routes

2.2 Hydrogen and Ammonia: Zero-Carbon Fuel Carriers

Alavi-Borazjani et al. (2025) explore Hydrogen and ammonia as promising zero-carbon alternatives for maritime propulsion, particularly for long-distance shipping and large vessels. Hydrogen offers the highest energy density among alternative fuels and produces zero carbon emissions when combusted in fuel cells or conventional engines. Its extremely low volumetric energy density requires cryogenic storage or compression, posing substantial onboard storage challenges (Jin et al., 2025). Paraschiv et al. (2025) identified green hydrogen produced from renewable electricity via water electrolysis as the most environmentally sustainable hydrogen pathway, though production costs remain significantly higher than those of conventional fuels.

According to Melnyk et al. (2025), Ammonia presents complementary advantages and challenges compared to hydrogen. The authors suggested that, owing to its superior energy density and lower storage requirements, ammonia can be stored at moderate pressures and temperatures, making it more compatible with existing maritime infrastructure (Melnyk et al., 2025). Nevertheless, ammonia's potential to generate harmful nitrogen oxides (NOx) during combustion poses significant operational and environmental concerns (Duong & Kang, 2025). Recent research by Imhoff et al. (2021) demonstrates that ammonia, when used in dual-fuel engines with proper combustion optimization, can achieve substantial reductions in emissions. The performance is optimized when blended appropriately with conventional fuels (Imhoff et al., 2021). Figure 1 shows the performance of alternative fuels and their contribution to emission reductions. Figure 1a) demonstrates the well-to-wake GHG Emissions Reduction, Figure 1b) energy density comparison, Figure 1c) project fuel cost, and Figure 1d) fuel storage capability.

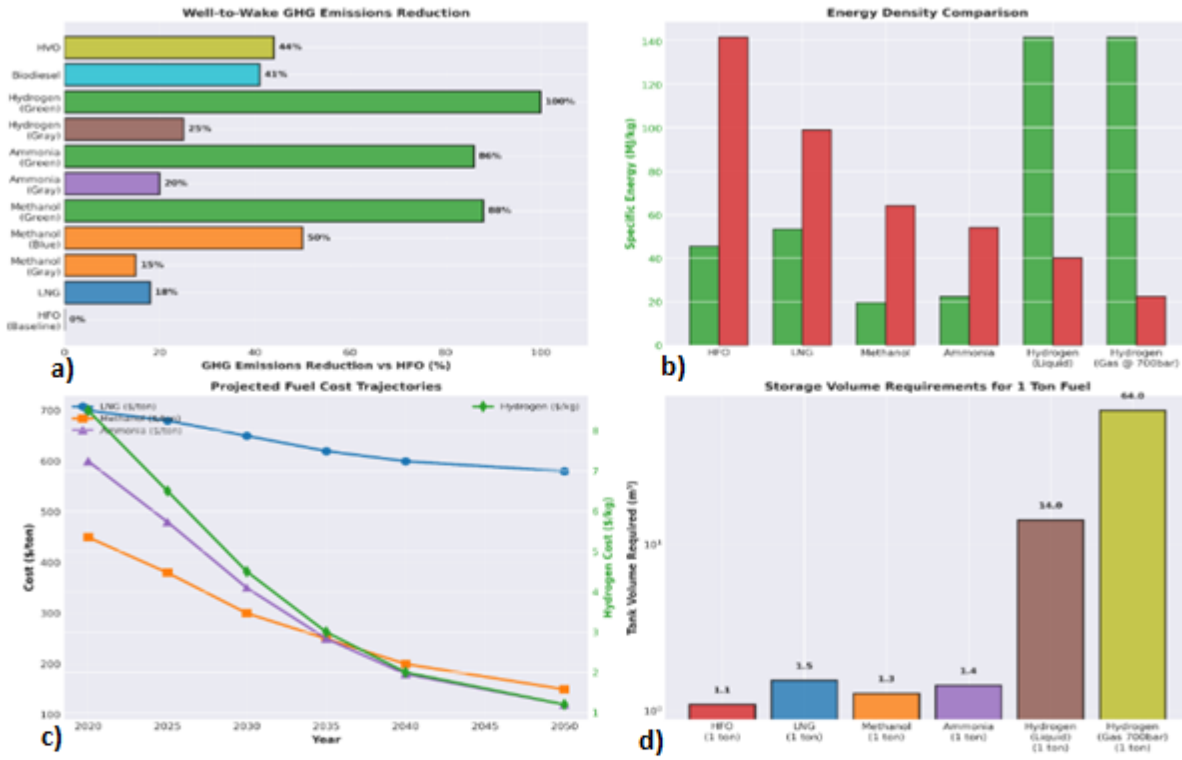

Figure 1. Alternative Fuels – Comprehensive Performance Analysis

Jin et al. (2025) conducted a comparative analysis of hydrogen and ammonia and found distinct applications based on vessel type and voyage distance. For short-sea shipping and localized operations, Lee et al. (2024) identified that hydrogen fuel cells offer superior environmental performance and operational efficiency. Conversely, ammonia's higher energy density and easier handling make it more practical for deep-sea applications, particularly given the infrastructure requirements for global adoption (Lee et al., 2024). Melnyk et al. (2025) carried out a comprehensive multicriteria assessment across technical, economic, and environmental dimensions. The result indicates that ammonia has a stronger potential for medium- and long-distance shipping due to lower transport costs and compatibility with existing liquefied gas infrastructure (Melnyk et al., 2025).

2.3 Methanol and Biofuels as Intermediate Solutions

Wang et al. (2025) show that Methanol is an emerging fuel type able to provide a significant intermediate solution for maritime decarbonization. Methanol can be produced via multiple pathways, including fossil-based gray methanol, carbon-neutral blue methanol with carbon capture, and renewable green methanol from biomass or electrolysis (Wang et al., 2025). A recent study by Lee et al. (2024) revealed that e-methanol and biomethanol can reduce GHG emissions by 88% and 92%, respectively, compared to conventional marine fuels. These fuel types maintain better compatibility with existing engine designs than hydrogen or ammonia (Lee et al., 2024). Elmallah et al. (2025) added that relatively simple engine modifications would be required for methanol operation. This should be coupled with established liquid-fuel bunkering practices to facilitate faster deployment than hydrogen or ammonia technologies (Elmallah et al., 2025).

Tay & Konovessis (2023) explore biofuels and biomethanol as alternative renewable fuels that leverage existing maritime fuel infrastructure and engine technologies. The authors specifically assessed biofuel impacts on marine diesel engines and demonstrated that advanced biofuels can achieve GHG emission reductions of 41.2% to 45.6% with minimal engine modifications. Although increased storage volumes (up to 23% greater than conventional fuel) present operational constraints (Otsason et al., 2025). The availability of sustainable biomass feedstocks remains a significant limitation for scaling biofuel adoption to meet global maritime demand, with current production capacity insufficient to satisfy more than a fraction of the industry's requirements (deManuel-Lpez et al., 2024).

3. Methodology

A systematic approach is adopted to explore the multifaceted advances in sustainable energy systems for marine operations. The study methodology explores the state of knowledge on alternative fuels, including LNG, methanol, ammonia, and hydrogen; energy efficiency technologies encompassing hull optimization, waste heat recovery, and wind assistance; electric and fuel cell propulsion systems; and digital technologies, including Internet of Things (IoT), artificial intelligence, and autonomous systems. The approach aimed to systematically capture the trend and identify areas for future research in energy efficiency and advanced propulsion technologies.

3.1 Hull Design and Hydrodynamic Optimization

Mallouppas & Yfantis (2021) identified that improving ship hull efficiency represents one of the most cost-effective and immediately implementable strategies for reducing fuel consumption and emissions. Advanced hull form optimization, employing computational fluid dynamics and artificial intelligence-driven design methodologies, can reduce resistance and fuel consumption by 8-15% compared to conventional hull designs (Tadros et al., 2023). Demirel et al. (2025) reported that Innovative technologies, including air lubrication systems, reduce friction through strategic bubble injection, achieving fuel savings of approximately 5-8% with relatively modest installation costs and operational complexity. Kawa (2025) added that surface treatment innovations, such as drag-reduction coatings and advanced hull cleaning systems, provide continuous benefits throughout a vessel's operational life. This prevents biofouling accumulation that can increase fuel consumption by up to 55% (Kawa, 2025).

Zhao et al. (2025) explore propeller design optimization by developing energy-saving devices and advanced blade designs. The study identified the design as another significant opportunity for efficiency gains in marine operations (Zhao et al., 2025). Further research by Huang et al. (2023) demonstrates that combined hull and propeller optimization can achieve cumulative fuel savings exceeding 20-25% while maintaining vessel performance and structural integrity. Hffmeier & Johanson (2021) added that the modular and retrofit-friendly nature of many advanced hull efficiency technologies should facilitate adoption across both new construction and existing vessel fleets. Although installation costs and operational disruptions require careful economic analysis (Hffmeier & Johanson, 2021).

3.2 Waste Heat Recovery and Advanced Thermodynamic Systems

Duong et al. (2022) revealed that waste heat generated during diesel engine operation accounts for approximately 40-50% of the fuel energy input. This offers a substantial opportunity for energy recovery and conversion to useful work in marine operations (Duong et al., 2022). Honcharuk (2025) identified that Organic Rankine Cycle (ORC) systems have proven highly effective for maritime waste heat recovery. The ORC can achieve fuel savings of 3-7% by converting low-temperature exhaust heat into additional propulsion power or electrical generation (Honcharuk, 2025). Zhao et al. (2025) simulation study revealed that combined wind-assisted propulsion and ORC waste heat recovery systems achieve maximum cumulative fuel savings of approximately 21% through synergistic optimization of both technologies. Polyvoda & Samoilov (2025) describe turbo-compounding systems that extract mechanical energy from exhaust gases via turbine-driven compressors as an alternative to waste heat recovery. This technology is particularly suited for high-power marine diesel engines, with implementation complexity offset by substantial efficiency gains (Polyvoda & Samoilov, 2025).

Law et al. (2023) integrated waste heat recovery with advanced cooling systems and auxiliary power management to comprehensively optimize onboard energy systems. Uyank et al. (2022) studied innovations in thermoelectric generators and advanced cryogenic technologies through the range of thermal energy sources for vessel operations. However, successful waste heat recovery implementation requires careful system integration considering engine operating conditions, vessel motion characteristics, and load variations across different voyage stages (Bucknall et al., 2018).

3.3 Wind-Assisted Propulsion and Renewable Energy Systems

Demirel et al. (2025) explore the modernization of wind-assisted propulsion systems (WAPS) that incorporate rigid sails, tractor kites, and rotor sails. The WAPS have advanced significantly from historical sailing technology to sophisticated automated systems (Demirel et al., 2025). Zhao et al. (2025) analyzed rotor sail technology based on Magnus-effect aerodynamics. The authors revealed that achievable wind-assisted fuel savings of approximately 8-12% depend on voyage route and wind conditions, with greater potential on specific trade routes that experience consistent favourable wind patterns (Zhao et al., 2025). Vidovi et al. (2023) noted that automated wing-sail and kite-assisted propulsion systems improve flexibility through modular designs, enabling retrofit installation on existing vessels. According to Demirel et al. (2025), comprehensive energy-efficiency assessments ranked wind-assisted propulsion among the highest-performing technologies, with respect to both the magnitude of fuel savings and cost-effectiveness.

Koumentakos (2019) studied the integration of solar energy on vessels. The author identified that solar energy currently provides modest direct propulsion contributions (3-7% of total power requirements for conventional vessels) and offers meaningful auxiliary power generation. Recently, solar energy integration has expanded significantly across specialized vessel types and shoreside port facilities (Koumentakos, 2019). Stamatakis & Ioannides (2021) explored the integration of photovoltaic systems with energy storage systems. The authors reveal that photovoltaic integration enables partial electrification of auxiliary systems, reduces fuel consumption for hotel loads during port operations and low-speed transits (Stamatakis & Ioannides, 2021). Laryea & Schiffauerova (2025) suggested that hybrid renewable energy systems (combining solar, wind, and battery storage with conventional propulsion systems) demonstrate substantial promise for near-coastal and short-sea operations. The integration could be further enhanced through energy management systems that leverage artificial intelligence to optimize power distribution (Laryea & Schiffauerova, 2025).

3.4 Slow Steaming and Operational Optimization Strategies

Mallouppas & Yfantis (2021) identified speed reduction as one of the most immediate and cost-effective strategies for emissions mitigation available to shipping operators. The authors added that reducing average operational speed by 10% results in approximately a 27% reduction in ship fuel consumption and associated emissions (Mallouppas & Yfantis, 2021). According to Orlandi et al. (2021), slow-steaming strategies, in combination with advanced voyage planning and real-time metocean data, reduce fuel consumption. By integrating artificial intelligence-driven weather-routing optimization, an additional 5-15% in fuel savings is achieved under balance distance, prevailing conditions, and operational constraints (Orlandi et al., 2021).

Viktorelius (2020) explored fleet-wide optimization strategies on operational efficiency of shipping networks, considering the integration of scheduling flexibility, demand management, and supply chain coordination. The authors identified that, through the integrated approach, remarkable efficiency improvements can be achieved across entire shipping networks (Viktorelius, 2020). Additionally, Motlagh et al. (2023) revealed that operational measures, trim optimization, ballast water management, and fouling prevention contribute incrementally to emissions reduction while requiring minimal capital investment. By implementing the Ship Energy Efficiency Management Plans (SEEMP) mandated by IMO (Issa et al., 2022), operational efficiency measures, fleet-wide optimization, and machine learning algorithms would help in optimizing predictive performance.

3.5 Digital Technologies and Smart Maritime Operations

Durlik et al. (2023) revealed that Internet of Things (IoT) technologies have revolutionized maritime operations by deploying sophisticated sensor networks that provide real-time monitoring of engine performance, hull conditions, weather impacts, and fuel consumption. The authors found that IoT-enabled vessel monitoring systems can collect terabytes of operational data daily, enabling predictive maintenance algorithms that reduce unplanned downtime by 40-50%. This integration enhances maintenance scheduling optimization and minimizes dry-dock time and associated costs (Joshva et al., 2024). A recent study by Vu et al. (2024) shows that real-time fuel consumption monitoring, combined with AI-driven optimization, enables vessel operators to achieve 3-5% fuel savings through continuous operational adjustments. Additionally, advanced sensor technologies, such as LiDAR, thermal imaging, and multispectral sensors, enhance maritime safety and environmental monitoring capabilities, particularly in challenging weather conditions and restricted visibility scenarios (Jacyna-Goda et al., 2025). Tay et al. (2021) emphasized that integrating data from multiple sensor sources, combined with machine learning algorithms for anomaly detection, enables early warning systems for equipment failure and supports maritime accident mitigation. Song (2021) added that the seamless integration of shore-based fleet management centers with onboard IoT systems enables remote decision support and optimized vessel performance across global shipping networks.

Recently, Alamoush & Ler (2025a) explored the application of Artificial Intelligence in maritime operations. The authors found that Artificial Intelligence applications in maritime operations span predictive fuel consumption modelling, optimal route planning, cargo optimization, and autonomous navigation support systems (Alamoush & Ler, 2025a). Nguyen et al. (2024) revealed that machine learning algorithms trained on massive historical operational datasets achieve remarkable accuracy in predicting fuel consumption across varying conditions. The authors found that the Random Forest and Tweedie regression models achieved R^2 values exceeding 0.99 (Nguyen et al., 2024). AI-powered route optimization systems that incorporate real-time metocean forecasts, vessel-specific performance characteristics, and optimal voyage plans reduce fuel consumption by 5-15% (Orlandi et al., 2021). Madusanka et al. (2023) confirmed that virtual replicas of physical vessels and port operations, synchronized with real-time operational data, enable comprehensive performance optimization and predictive maintenance. Such synchronization can support port digital twins by incorporating vessel dynamics, cargo flows, equipment performance, and environmental conditions, thereby reducing waiting times by 75% and associated emissions from vessel idling (Eom et al., 2023. Advanced multi-physics modelling of vessel operations enables testing of alternative fuel systems, energy-efficiency retrofits, and autonomous navigation capabilities before implementation, reducing retrofit risks and accelerating the adoption of proven technologies (Alamoush & Ler, 2025a).

Furthermore, Alamoush & Ler (2025b) present that the Maritime Autonomous Surface Ships (MASS) is a transformative technology with the potential to optimize vessel operations, improve safety, and reduce crewing requirements. The authors added that Level 1 and 2 autonomous systems would enable remote ship operation and support automated decision-making. With an improved development strategy, the MASS is advancing toward commercialization, with positive trials conducted by leading shipping companies and technology providers (Arnaoot, 2025). Joshva et al. (2024) added that autonomous navigation systems eliminate human error from steering, navigation planning, and collision avoidance. This strategy demonstrates potential 5-20% fuel efficiency improvements through optimized speed and route profiles (Joshva et al., 2024). The regulatory framework governing the deployment of autonomous vessels remains nascent, with the IMO developing guidelines and class societies establishing certification procedures (Alamoush & Ler, 2025b). Nganga et al. (2023) explore remote operations center (ROC) strategies, in which trained operators supervise multiple autonomous vessels from shore-based facilities. The study examines present substantial opportunities for operational optimization and immediate emergency response capabilities (Nganga et al., 2023). Tabish & Chaur-Luh (2024) identified security challenges, such as cybersecurity threats and system reliability requirements, as crucial in MASS operation and suggested multilayered protection systems and dedicated cybersecurity operations centers for maritime applications.

4. Discussion and Future Investigation Priorities

Despite substantial research progress, significant knowledge gaps persist regarding long-term operational performance and environmental credibility of emerging maritime technologies under diverse real-world conditions (Wang et al., 2025). Comparative lifecycle assessments across alternative fuel production pathways incorporating evolving electricity grid compositions and emerging production technologies require ongoing refinement as energy systems decarbonize ((Stockbruegger & Bueger, 2024; Wang et al., 2025). Holistic assessment frameworks that integrate the technical, economic, safety, social, and environmental dimensions of sustainable energy technologies remain underdeveloped and provide gaps for future research.

The interactive effects and synergies among multiple technologies merit deeper investigation, as coordinated implementation of complementary solutions offers greater potential for emissions reduction than individual technologies deployed in isolation (Zhao et al., 2025). Safety and reliability assessments for emerging technologies, particularly autonomous systems and advanced fuel-handling procedures, require accelerated research and real-

https://doi.org/10.17605/OSF.IO/2PBT7

world testing to build regulatory confidence and operational viability (DiVaio et al., 2025; Liu et al., 2024). The distributional implications of maritime decarbonization across different stakeholder groups, including seafarers, port workers, and developing-nation maritime industries, require sustained and integrated research to inform equitable transition pathways.

6. Conclusion

The maritime industry stands at a critical juncture where technological innovation, policy frameworks, and economic mechanisms must converge to achieve ambitious climate goals while maintaining the sector's fundamental role in global trade and economic development. Substantial advances in alternative fuels, energy efficiency technologies, and digital systems demonstrate that comprehensive decarbonization is technically feasible and increasingly economically competitive with conventional approaches (Uchechukwu & Mba, 2024). Hydrogen, ammonia, and advanced biofuels offer long-term pathways toward zero-carbon maritime operations, while LNG and methanol provide bridging solutions enabling near-term emissions reductions (Mallouppas & Yfantis, 2021).

Beyond fuel transitions, systematic improvements in hull design, propulsion efficiency, and operational practices generate immediate emissions reductions achievable through near-term implementation (Tadros et al., 2023). Digital technologies and artificial intelligence enable unprecedented opportunities for operational optimization, while autonomous systems and advanced monitoring capabilities enhance both efficiency and safety (Joshva et al., 2024). However, realizing this technological potential requires coordinated policy action that establishes clear regulatory signals, removes infrastructure barriers, and provides strategic investment in emerging technologies (Ibokette et al., 2024).

The IMO 2050 decarbonization target, while ambitious, remains achievable through an integrated strategy combining multiple technological approaches deployed across diverse maritime contexts (Halim et al., 2018). Successful transition necessarily involves substantial capital investment, workforce development, and international cooperation, yet the environmental and economic benefits of avoided climate change impacts far exceed transition costs (LingChin et al., 2024). The maritime industry's decarbonization journey will serve as a model for other transport sectors facing similar sustainability challenges, demonstrating that global decarbonization, while complex, remains within the realm of practical possibility with sufficient commitment, innovation, and cooperation.

Conflict of Interest

The authors declare that they have no competing interests

Funding Statement

This research received no funding

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